



GIA - IAA

Introducció a

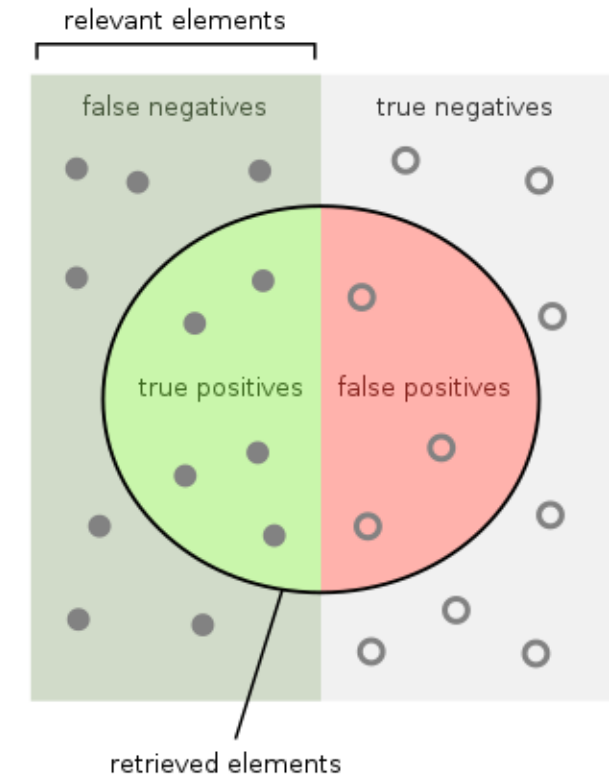
l'Aprenentatge Automàtic

6. Lab: Classification (II) Data imbalance

Jordi Luque
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Performance Metrics

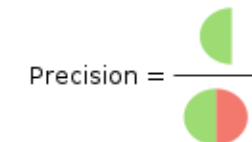
		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total population = P + N		
	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection



$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2tp}{2tp + fp + fn}.$$

https://en.wikipedia.org/wiki/Precision_and_recall

How many retrieved items are relevant?



How many relevant items are retrieved?



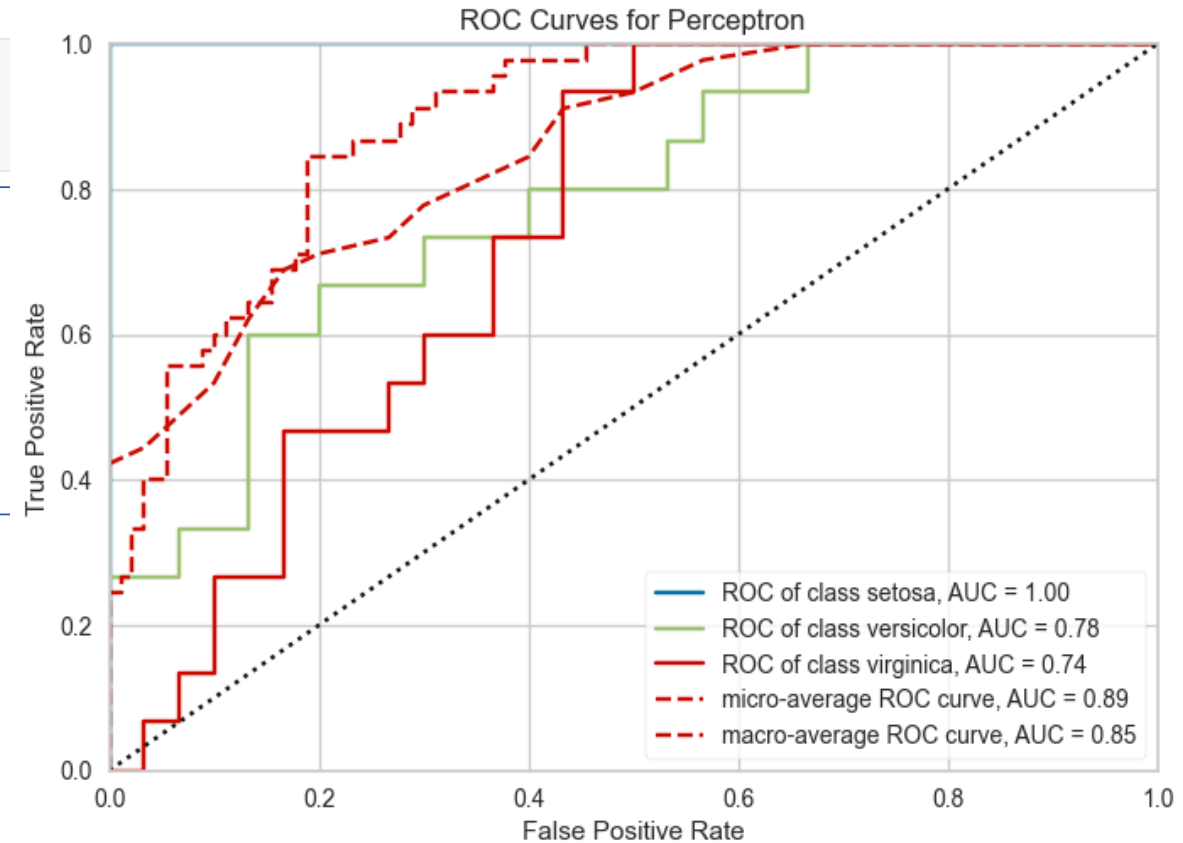
Performance metrics

```
1 print(classification_report(ppn.predict(X_test_std), y_test, target_names=iris.target_names))
2 results_df = save_results(ppn, X_test_std, y_test, 'Perceptron', results_df)
```

✓ 0.0s

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	15
versicolor	0.13	0.67	0.22	3
virginica	0.93	0.52	0.67	27
accuracy			0.69	45
macro avg	0.69	0.73	0.63	45
weighted avg	0.90	0.69	0.75	45

❖ Why accuracy is not a good performance metric under data imbalance?



Receiver Operating Characteristic curve

The dataset: Porto Seguro safe driver prediction

The screenshot shows the Kaggle website interface. On the left is a sidebar with navigation links: Home, Competitions (highlighted), Datasets, Models, Code, Discussions, Learn, and More. The main content area features a search bar, 'Sign In', and 'Register' buttons. A featured banner for the 'Porto Seguro's Safe Driver Prediction' competition is displayed, showing a prize of \$25,000 and the goal to 'Predict if a driver will file an insurance claim next year.' Below the banner are tabs for Overview, Data, Code, Models, Discussion, Leaderboard, and Rules. The 'Overview' tab is active, showing a timeline from 'Start' (Sep 29, 2017) to 'Close' (Nov 30, 2017), with a 'Merger & Entry' point. On the right, details about the 'Competition Host' (Porto Seguro), 'Prizes & Awards' (\$25,000), and 'Participation' are listed.

Competition Overview:

- Competition Name:** Porto Seguro's Safe Driver Prediction
- Prize Money:** \$25,000
- Goal:** Predict if a driver will file an insurance claim next year.
- Host:** Porto Seguro
- Prizes & Awards:** \$25,000, Awards Points & Medals
- Timeline:** Start (Sep 29, 2017) to Close (Nov 30, 2017), including Merger & Entry.

Download the data from:

<https://www.kaggle.com/c/porto-seguro-safe-driver-prediction>

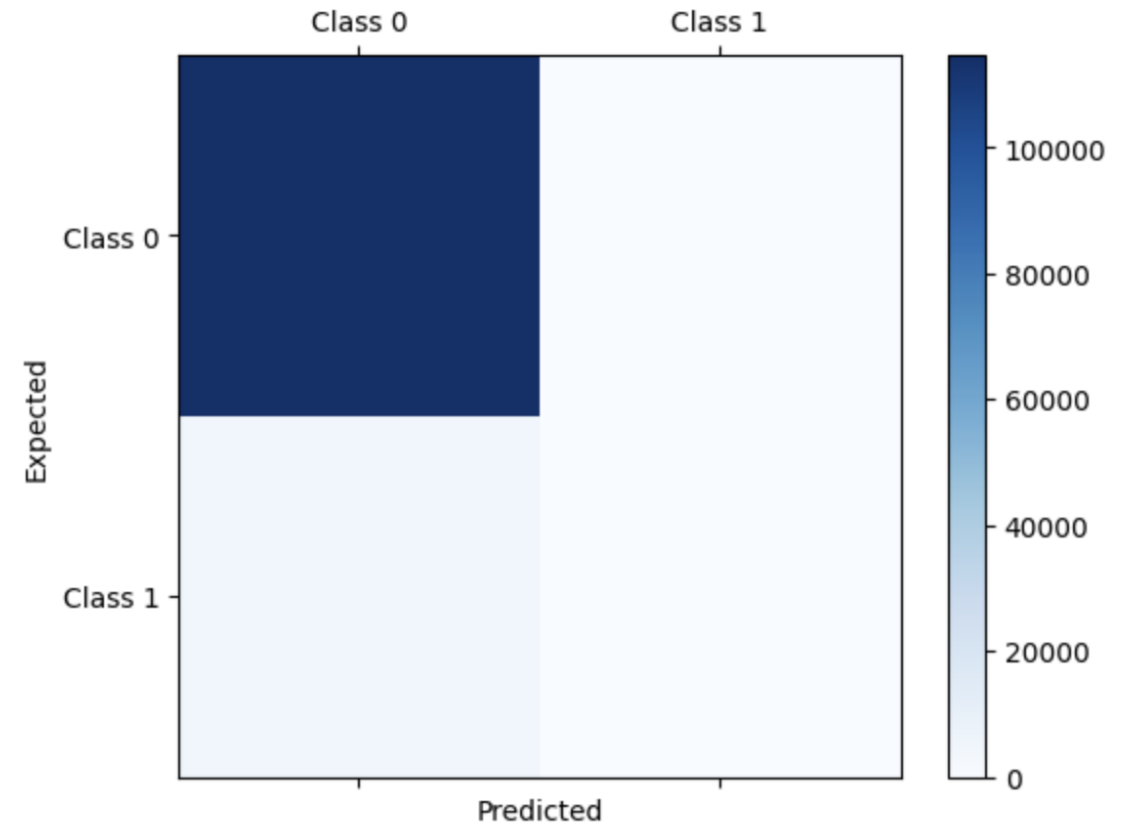
Performance metrics

```
1 print(classification_report(model.predict(X_test), y_test))
```

[54] ✓ 0.1s

	precision	recall	f1-score	support
0	1.00	0.96	0.98	119043
1	0.00	0.00	0.00	0
accuracy			0.96	119043
macro avg	0.50	0.48	0.49	119043
weighted avg	1.00	0.96	0.98	119043

❖ Why accuracy is not a good performance metric under data imbalance?



Performance metrics

```
1 print(classification_report(model.predict(X_test), y_test))
```

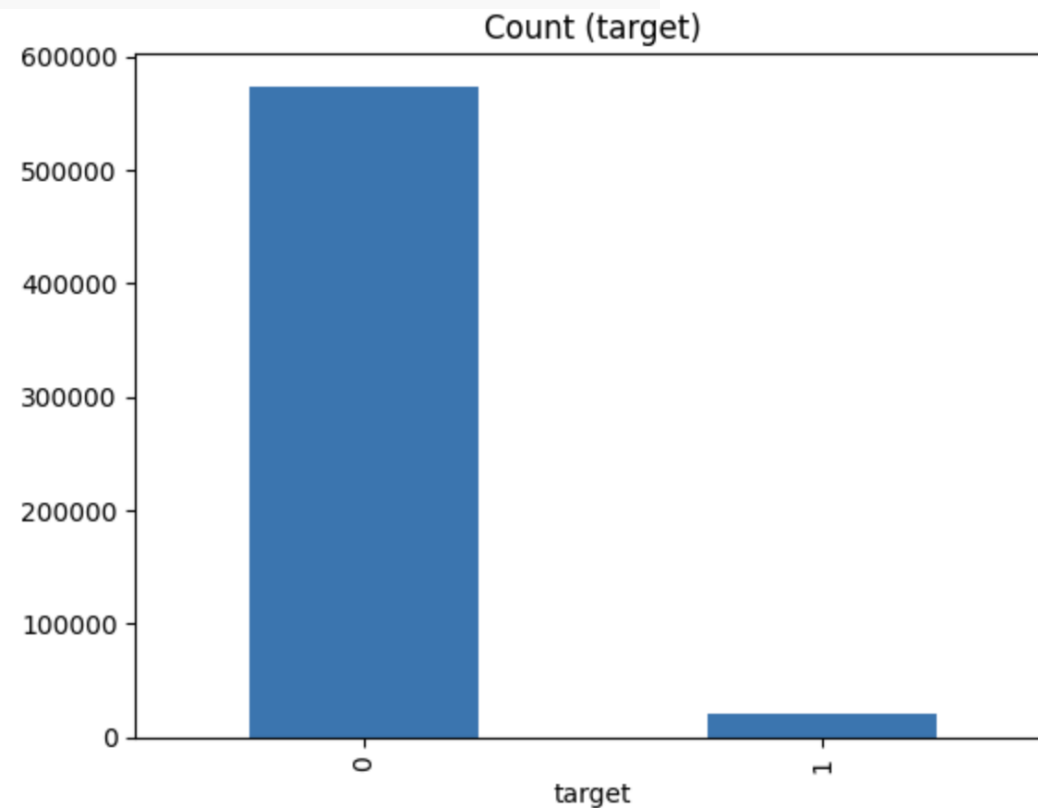
[54] ✓ 0.1s

	precision	recall	f1-score	support
0	1.00	0.96	0.98	119043
1	0.00	0.00	0.00	0
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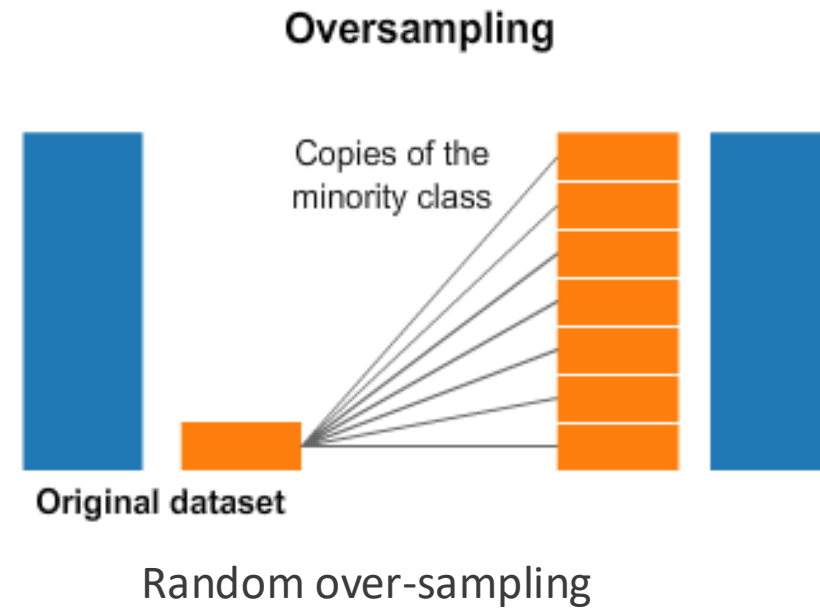
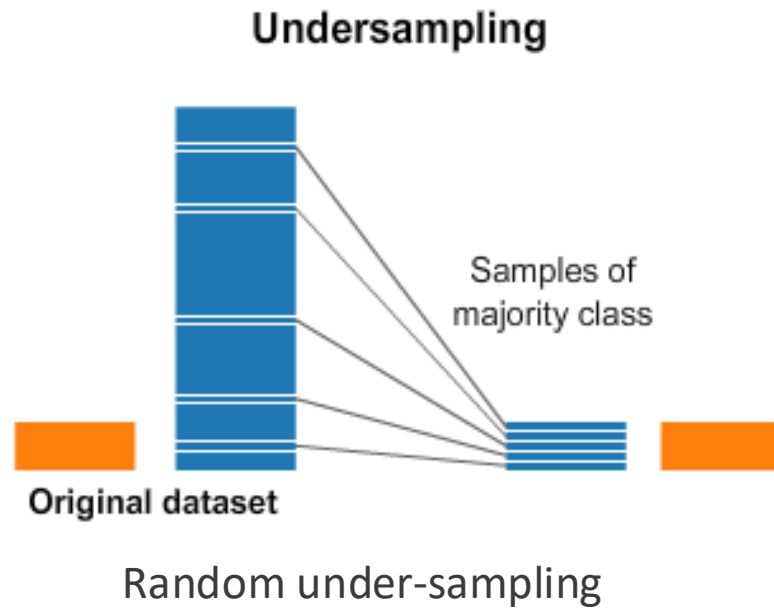
❖ Why accuracy is not a good performance metric under data imbalance?

Use `class_weight`

Most of the models in `scikit-learn` have a parameter `class_weight`. This parameter will affect the computation of the loss in linear model or the criterion in the tree-based model to penalize differently a false classification from the minority and majority class. We can set `class_weight="balanced"` such that the weight applied is inversely proportional to the class frequency. We test this parametrization in both linear model and tree-based model.



Resampling methods



- ❖ Study the performance in the dummy example using both techniques? Do you have any other ideas to balance the datasets?

The datasets (II)

ADULT dataset

```
1 from sklearn.datasets import fetch_openml
2 |
3 df, y = fetch_openml("adult", version=2, as_frame=True, return_X_y=True)
4 df = df.drop(columns=["fnlwgt", "education-num"])
```

[59] ✓ 0.6s

The screenshot shows the OpenML website interface. On the left is a dark sidebar with navigation links: Datasets (5.4k), Tasks, Flows, Runs, Collections, Benchmarks, Task Types, Measures, Learn, Documentation, Blog, API's, and Contribute. The main content area is titled 'adult' and includes the following information:

- ID: 1590, verified, ARFF, Public, 2015-06-09
- Joaquin Vanschoren, 0 likes, 0 issues, 0 downloads
- OpenML-CC18, OpenML100

At the top right of the main area are 'Sign In' and 'Sign Up' buttons. Below the dataset title, there are tabs for 'Data Detail', 'Analysis', and 'Tasks'. On the right side of the main area, there are icons for 'download', 'edit', 'json', and 'xml'. The 'Description' section contains the following text:

Author: Ronny Kohavi and Barry Becker
Source: UCI - 1996
Please cite: Ron Kohavi, "Scaling Up the Accuracy of Naive-Bayes Classifiers: a Decision-Tree Hybrid", Proceedings of the Second International Conference on Knowledge Discovery and Data Mining, 1996

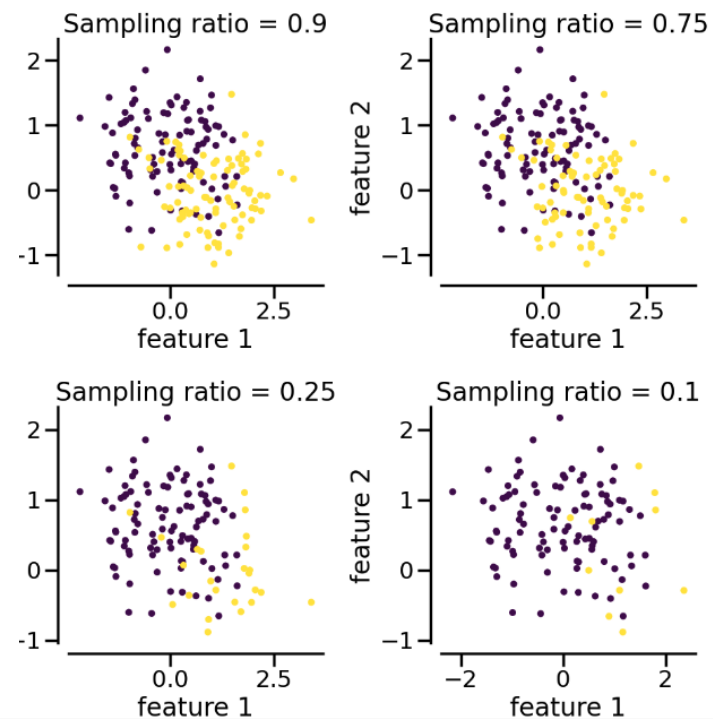
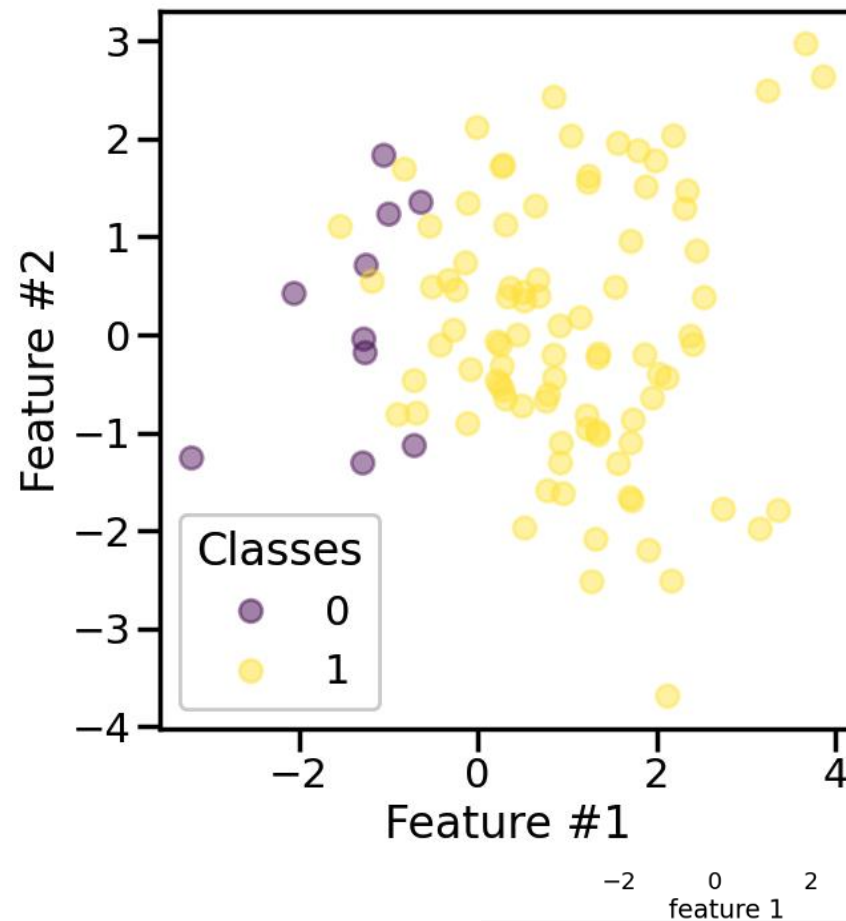
Prediction task is to determine whether a person makes over 50K a year. Extraction was done by Barry Becker from the 1994 Census database. A set of reasonably clean records was extracted using the following conditions: ((AAGE>16) && (AGI>100) && (AFNLWGT>1)&& (HRSWK>0))

This is the original version from the UCI repository, with training and test sets merged.

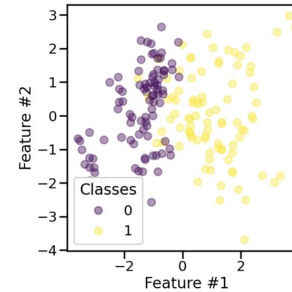
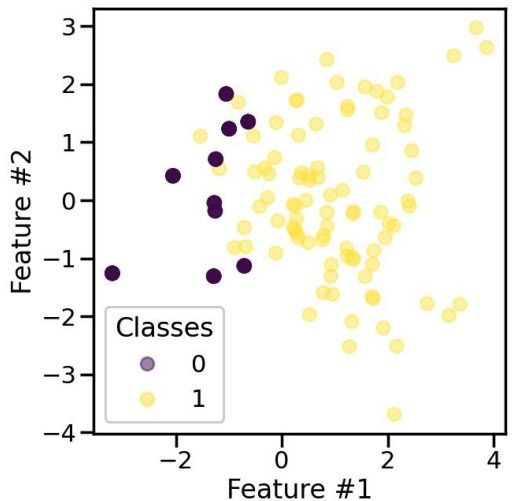
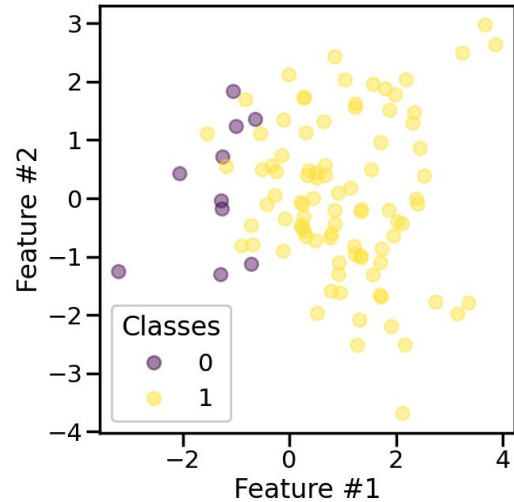
Python imbalanced-learn module

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3 from sklearn.datasets import make_moons
4
5 X, y = make_moons(n_samples=200, shuffle=True, noise=0.05)
6 X = pd.DataFrame(X, columns=["feature 1", "feature 2"])
7 ax = X.plot.scatter(
8     x="feature 1",
9     y="feature 2",
10    c=y,
11    colormap="viridis",
12    colorbar=False,
13 )
14 sns.despine(ax=ax, offset=10)
15 plt.tight_layout()
```

[49]



Shrinkage parameter in over-sampling



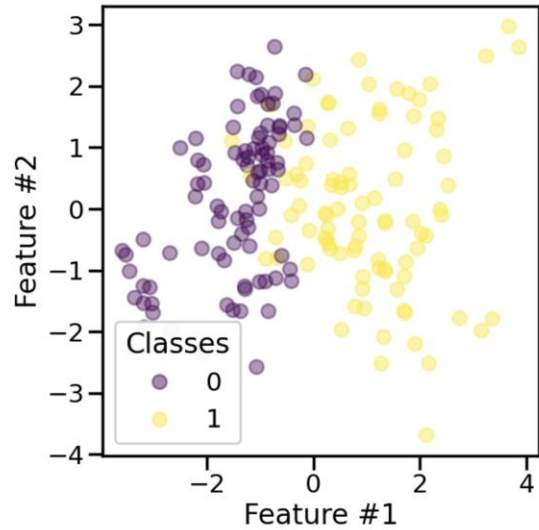
We can set **shrinkage** to a floating value to add a small perturbation to the samples created and therefore create a smoothed bootstrap.

```
1 sampler = RandomOverSampler(shrinkage=1, random_state=0)
2 X_res, y_res = sampler.fit_resample(X, y)
3 Counter(y_res)
```

[64] ✓ 0.0s

- ❖ Study the effect of the shrinkage parameter. Why the samples in class 0 are darker when we use over-sampling? What happens for higher values of the shrinkage?

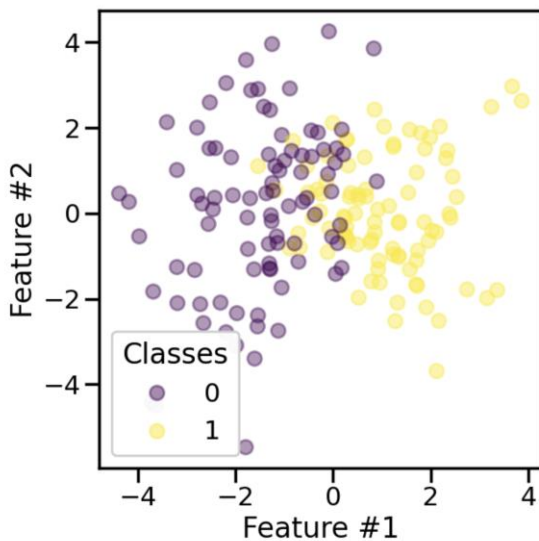
Shrinkage parameter in over-sampling



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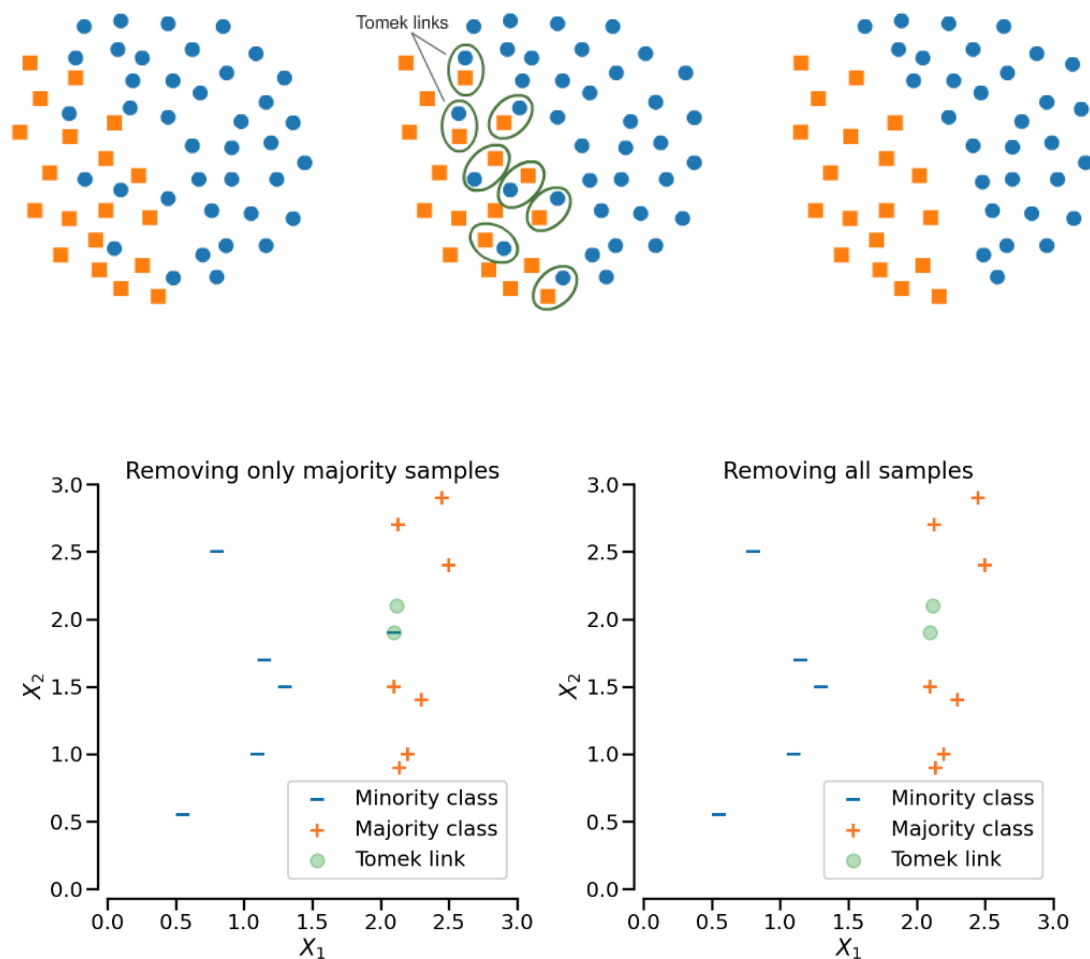
[64] ✓ 0.0s



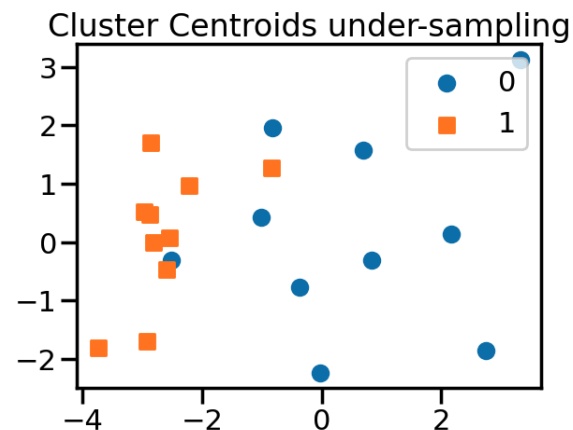
- ❖ Study the effect of the shrinkage parameter. Why the samples in class 0 are darker when we use over-sampling? What happens for higher values of the shrinkage?

Under-sampling methods: Tomek links and Cluster centroids

Tomek links

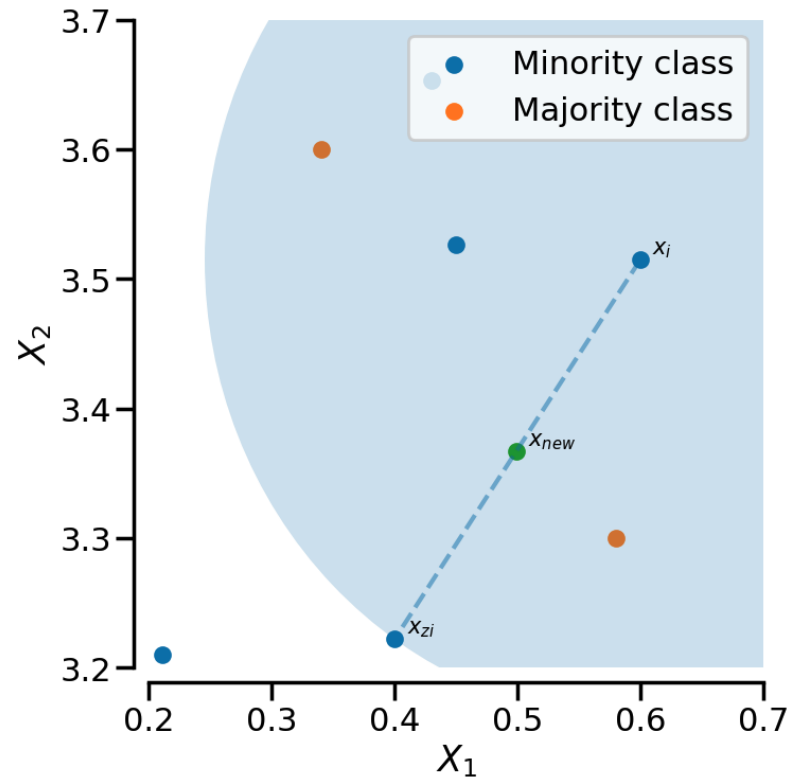


Cluster Centroids

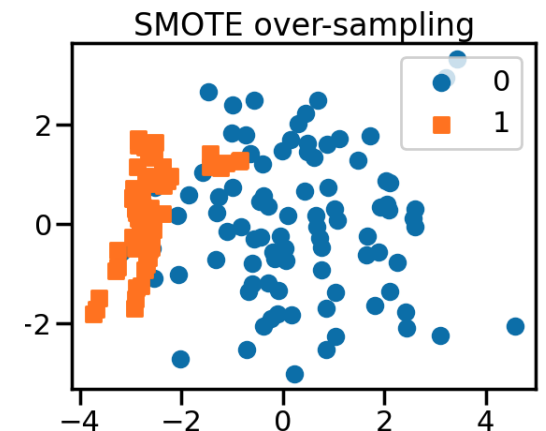
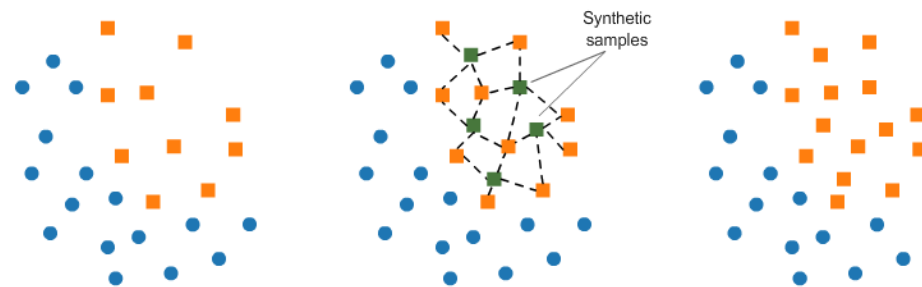


- Tomek links are pairs of very close instances, but of opposite classes. Removing the instances of the majority class of each pair increases the space between the two classes, facilitating the classification process.
- Cluster Centroids performs under-sampling by generating centroids based on clustering methods. The data will be previously grouped by similarity, in order to preserve information.

Over-sampling methods: SMOTE

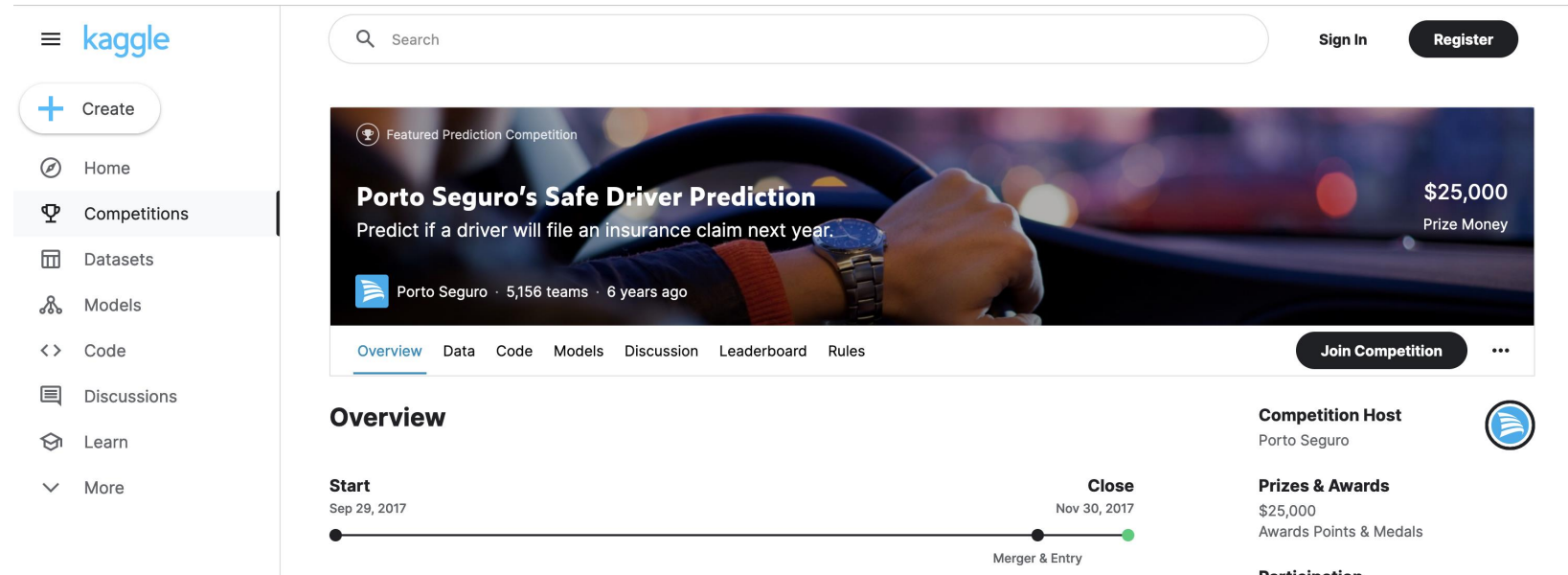


- SMOTE (Synthetic Minority Oversampling TEchnique) consists of synthesizing elements for the minority class, based on those that already exist. It works randomly picking a point from the minority class and computing the k-nearest neighbors for this point. The synthetic points are added between the chosen point and its neighbors.



- ❖ Study the different methods for under-sampling: random totem links, cluster centroids and over-sampling: random and SMOTE for the dataset proposed: [Porto Seguro safe driver prediction](#)

Dataset: Porto Seguro safe driver prediction



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<https://www.kaggle.com/c/porto-seguro-safe-driver-prediction>

- ❖ Study the different methods for under-sampling: random tomes links, cluster centroids and over-sampling: random and SMOTE for the dataset proposed. Plot performance report and a ROC curve for all the methods using Logistic Regression, SVMs, Decision Trees and KNN classifiers.